Statistical Design of Investigations

The Effect of Invitation Message Style on Survey Response Rates and Times

Experiment design and analysis report – ICA1

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Abstract

This experiment addresses the challenge of low response rates in online questionnaires by investigating how message personalization and friendship closeness affect response rates and times. A randomized 2x2 factorial design was used, with 138 participants assigned to receive either a basic or a personalized message. 66 of these participants were close friends and 72 were not-so-close friends. The constraints due to the skewness of the data resulted in a change of direction toward alternative methods other than ANOVA, such as GLMs. The results showed that friendship closeness had a significant effect on the response rate, contrary to message personalization. The high variance of the data made it impossible to distinguish any effects of the factors in the response time. To obtain better results when examining message personalization and friendship closeness, future studies should include larger sample sizes within a more homogeneous group of participants, handle right-skewed data using alternative methods, and conduct the experiment under more regulated conditions.

1. Introduction

Online questionnaires have become increasingly prevalent due to their benefits, which include low cost, scalability, and reduced transmission time. However, one of the drawbacks of these research instruments is that they suffer from low response rates.

According to previous studies, modifying invitation or reminder messages may have an impact on questionnaire response rates. For instance, personalizing salutations increase response rates by almost 40% compared to an impersonal salutation (Joinson & Reips, 2007), while non-informative content can better attract customer's attention and increase the possibility that they will read and respond to an advertisement message (Sahni et al., 2018).

This experiment aims to investigate whether the invitation message style influences online survey response rates and response times among friends.

2. Design and Data

2.1. Experimental Design

A brief 10 question survey (Appendix A) was sent with different invitation message styles, to investigate how the response behavior would change. The survey was intentionally concise to mitigate the impact of survey duration on response accuracy and participation. The survey content was identical for all participants to isolate the effect of personalization. Participants were individuals aged 20-26.

A 2x2 factorial design was used in this study. Factors included message style (Basic and Personalized Message levels) and friendship closeness (Close and Not-so-close levels), to account for the different relationship effect with the participants. The response variables being the response time and rate. 'Basic' and 'Not-so-close' were treated as the 'Low' levels of the factors. Thanks to the factorial structure, it was possible to efficiently examine their main effects and their interaction.

Participants were randomized to receive one of two message styles (Appendix B). Randomization was conducted to ensure a fair distribution across closeness levels, prevent selection bias, and verify that observed effects could be attributed to the intervention rather than confounding factors.

2.2. Data Collection

Messages were sent using a code-generated WhatsApp messaging system from Spain and Turkey at 20:30 local time on a Thursday, owing to the researchers' locations in these countries. To ensure participant comprehension while preserving internal validity, messages and Google Forms were standardized across 3 different languages: Spanish, Turkish, and English. Only the answers submitted before 23:59 next Sunday were taken into account for the analysis. The code recorded when messages were sent automatically.

Additionally, completion times were extracted from the Google Forms spreadsheet. In order to preserve participant anonymity, each person was given a unique participant ID.

2.3. Sample Size

In order to decide the necessary sample size for the experiment, Cohen's f^2 statistic was used, which measures the ratio of explained to unexplained variance (Cohen, 1988). Specifying a significance of $\alpha = 0.05$, a power of $\beta = 0.8$ and $f = 0.25$ (medium value according to (Cohen, 1988)) a sample size of 128 was obtained, equalling 32 people per factor combination.

In total, 138 individuals were tested, 66 of them being close friends and 72 being not-so-close friends.

2.4. Recorded Variables and Outcome Measures

For each recipient, 2 timestamps were recorded:

- **TimeSent**: The time invitation message was sent
- **TimeSurvey**: The time the recipient responded to the Google Forms

Response time was calculated as the time between when the message was sent and survey completion. Only the individuals who replied were subjected to response-time analysis.

3. Methodology

Following data collection, various statistical methods were employed to evaluate how message type and closeness influenced response rates and latencies.

3.1. Exploratory Analysis

An exploratory analysis was carried out to asses the data distribution and validity, and extract some initial conclusions from the data.

3.2. Significance Tests

Following the exploratory phase, statistical significance tests were carried out on the complete dataset to evaluate the main effects of Message Type and Friend Category, as well as the interaction between them.

3.2.1. Response Rate

Pearson's Chi-Square Test of Independence, a non-parametric test that does not make assumptions about the data distribution, was used to test for effects on the response rate by the importance factors. Logistic regression was used to also test for the interaction effects.

3.2.2. Response Time

As the experiment was designed as a 2x2 factorial design, a Two-Way Fixed-Effects ANOVA (Analysis of Variance) was used to test for differences in Response Time depending on the investigated factors. A fixed-effects model was chosen because of a lack of interest in how the results may generalize to other levels of the factors. In addition, diverse techniques were used to validate ANOVA assumptions (normality of residuals and homoscedasticity).

3.2.3. Normality Transformations

In the exploratory analysis, data was found to be right-skewed, so different transformations were used on the dataset, trying to normalize it so ANOVA assumptions could be fulfilled: Logarithmic transformation and Box-Cox transformation.

3.3. Division into two Days

Exploratory analysis showed a bimodal distribution, suggesting different response behaviors for the first day and the following days. The analysis of each mode separately was carried out by dividing the data into First Day data and Non-First Day data.

3.4. Other Methods: GLMs

As ANOVA assumptions were violated in every experiment, Generalised Linear Models (GLMs) were found to be well-suited for right skewed data, such as reaction time (Lo & Andrews, 2015). This type of model does not assume that the error terms are normally distributed, allowing the response variable (response time) to follow any distribution from the exponential family. Gamma and Quasi-Poisson components were used to test for statistically significant differences in the response latency.

4. Results

4.1. Exploratory Analysis

In the grouped kernel density plot (Figure 1) it can be seen that the data is skewed to the right, as expected with time data. Close friends show a greater concentration of answers in the first minutes, as well as a second peak after 1000 minutes, which corresponds to the morning on the second day.

The interaction plot (Figure 2) shows how both being a close friend and receiving a personalized message reduce the mean response time, with the reduction in personalized messages being lower for close friends, although the variance of the data is very high.

Due to right-skewness, we repeated this analysis after applying a logarithmic transform, aiming to achieve normality (Figure 3). However, the multimodality of the data causes the logarithmic transform to not result in normally distributed data.

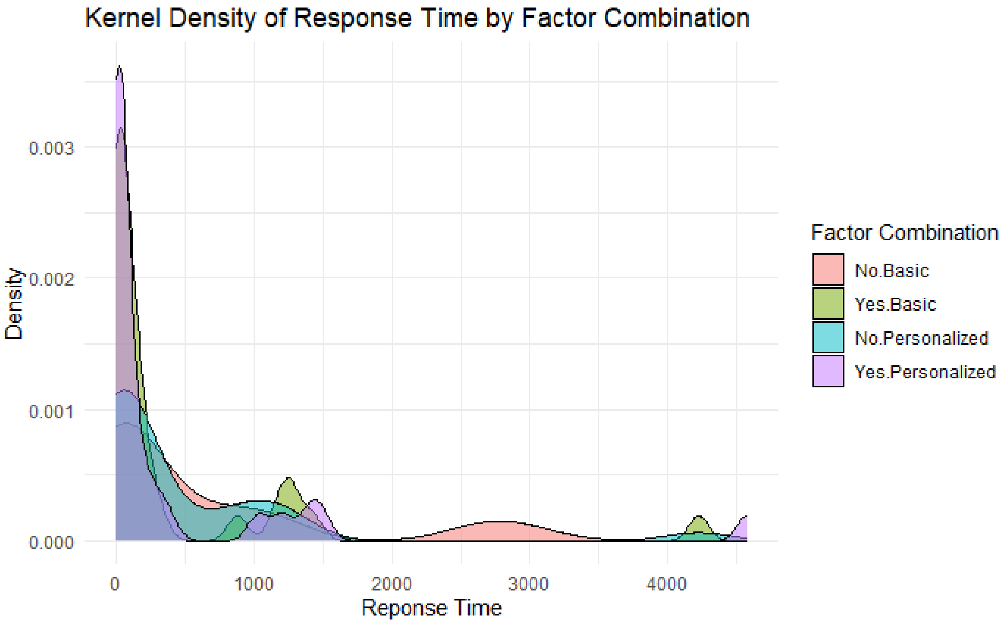

Figure 1: Kernel density estimates of ResponseTime across four factor combinations.

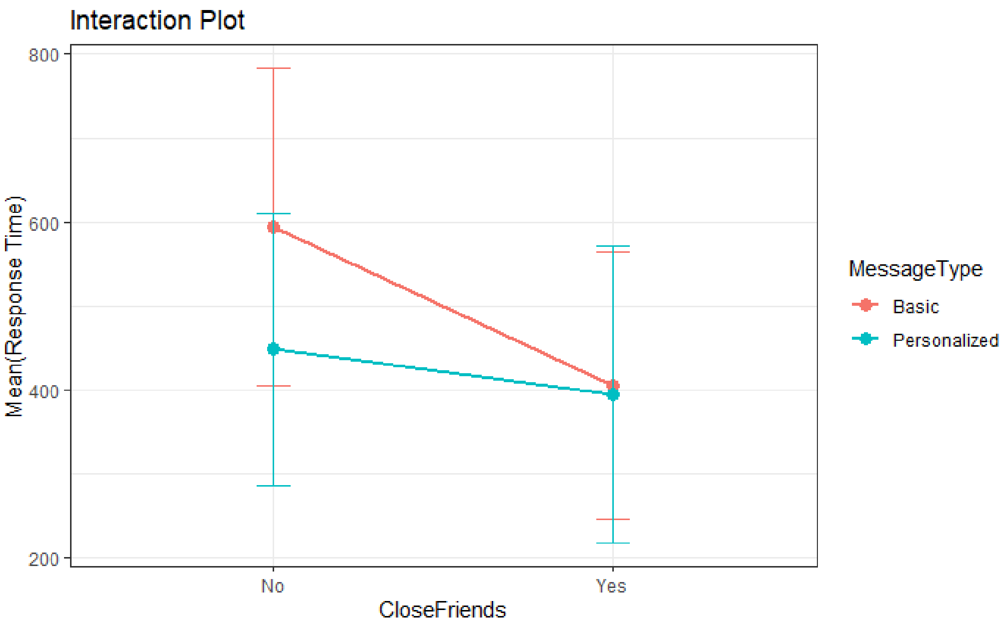

Figure 2: Interaction plot to display the differences in the means for each factor combination.

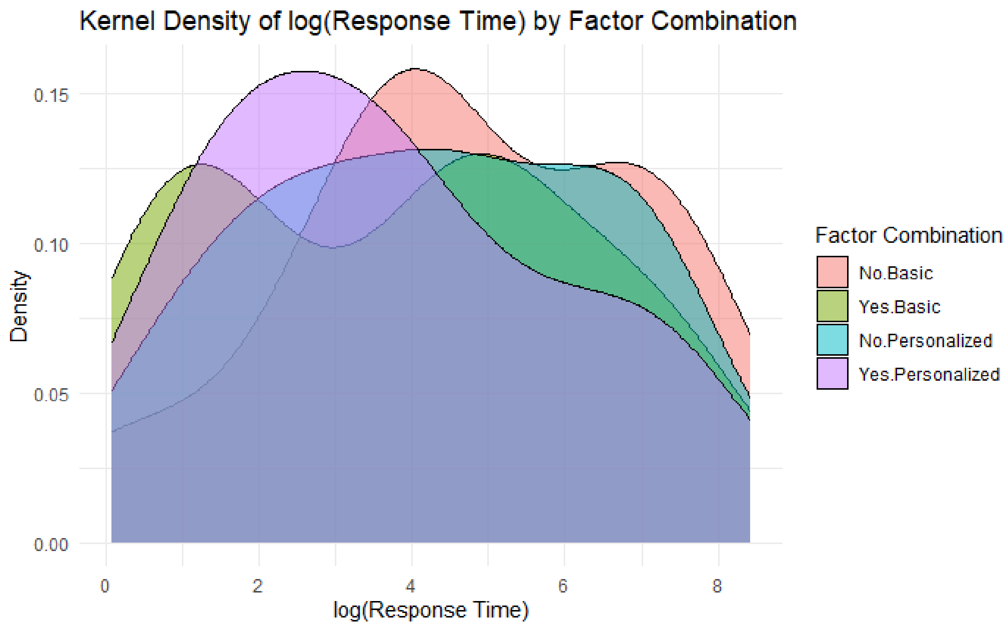

Figure 3: Kernel density estimates of log(ResponseTime) across four factor combinations.

4.2. Significance Tests

4.2.1. Response Rate

Neither MessageType ($p = 0.676$) nor CloseFriends ($p = 0.067$) had a statistically significant effect on the response rate according to the Chi-Square test.

Logistic regression was also used to test the effect of both variables and their interaction. This model found closeness to have a significant effect ($p = 0.044$), but not personalization nor interaction. Results from this analysis are shown in Table 1.

These results suggest a significant relationship between closeness and response rates, although the Chi-square test only approached the conventional significance threshold.

Factor	Level	Answered	Not Answered	Odds Ratio (OR)	95% CI
Friendship Status	Not Close Friend	52	20	3.625	[1.034,12.705]
	Close Friend	57	9		
Message Style	Basic Message	53	16	1.750	[0.614,4.989]
	Personalized Message	56	13		
Interaction				0.441	[0.0756,2.564]

Table 1: Response rates per factor level and results of the logistic regression.

4.2.2. Response Time

The ANOVA analysis was first performed on the original dataset, but the normality assumptions were violated, as expected. A logarithmic transform was then performed on the data, but the residuals still were not normal. A Shapiro-Wilk test was performed to confirm this ($p = 0.021$).

A BoxCox transform was also used, but the optimal lambda obtained was close to zero, so its effect was very similar to that of the logarithmic transform.

4.3. Division into two Days

Based on the exploratory analysis, the boundary between the first and second day was set at 500 minutes (4:50am of the second day), as there was a gap in responses from 370 to 510 minutes.

4.3.1. Response Rate

The effect of closeness in the first day response rate was found to be significant ($p = 0.019$) in the chi-square test. However, no significant difference was found between message types ($p = 0.489$).

Logistic regression was also performed, with only CloseFriends having a significant effect ($p = 0.037$). Results from this analysis are shown in Table 2.

Factor	Level	Answered	Not Answered	Odds Ratio (OR)	95% CI
Friendship Status	Not Close Friend	37	35	3.220	[1.190,8.710]
	Close Friend	46	20		
Message Style	Basic Message	38	31	1.750	[0.688,4.450]
	Personalized Message	43	26		
Interaction				0.571	[0.140,2.328]

Table 2: First-day response rates per factor level and results of the logistic regression.

4.3.2. Response Time

ANOVA analysis was performed on the response time of the participants who answered on the first day, but the normality assumptions were still violated, even with a logarithmic transformation in the Shapiro-Wilk test ($p = 0.019$).

4.4. Other Methods

4.4.1. GLMs

The data was fitted to a Gamma GLM with a logarithmic link function, accounting for the right-skewedness of the data. The analysis revealed that neither the effects of CloseFriends ($p = 0.515$), MessageType ($p = 0.695$), nor their interaction ($p = 0.745$), reached statistical significance. Furthermore, the model showed very low predictive power, explaining only 0.63% of the total deviance ($D^2 = 0.63\%$)

A Quasi-Poisson GLM was also used, but the model still failed to explain the variance of the data, with $D^2 = 1.13\%$ and p-values over 0.5 for all variables.

4.5. Sample Size Analysis

The sample size was too small to find significant evidence due to a higher variance than expected. A posterior sample size analysis was performed given the results of the Gamma GLM. A value of $f^2 = 0.0063$ was obtained for Cohen's statistic based on Equation 1.

$$f^2 = 1 - \frac{D_{\text{residual}}}{D_{\text{null}}} \tag{1}$$

Given this value, a sample size of 1242 was found to be necessary to obtain a power of $\beta = 0.8$ and a significance of $\alpha = 0.05$ given the variance of the data.

5. Discussion

The analysis indicates that closeness significantly impacted engagement, especially in the first day. However, message personalization showed no detectable effect on participation. Furthermore, neither social closeness nor personalization had a significant effect on the response latency, although a slight reduction in the mean average time to respond was seen in both closeness and personalization in the interaction plot.

Normalization transformations failed because of the bimodal distribution of the data, which was characterized by first day and delayed responses with a nighttime gap. Consequently, the study transitioned from the initially planned ANOVA analysis to GLMs, which do not assume normality. This adjustment allowed for the drawn conclusions to be statistically sound despite the lack of normality.

However, GLMs poorly explained the deviance in the data, as the used components assume unimodality in the data distribution. The high p-values also suggest that the experimental factors accounted for a negligible portion of the variance in response times.

These results can be explained by the nature of the participants. Close friends are obviously more eager to help out in a short survey. However, as friends are already keen on helping each other, personalization has no effect on their willingness to respond, contrary to previous studies within the general population. The high variance of the data made it impossible to distinguish any effects on the response latency.

5.1. Further Improvements

Despite our careful design, the analysis highlighted some areas that could have been improved.

The limited resources (number of friends) could have been used more efficiently. The survey required very little effort (< 1 minute), so participants filled it regardless of the message personalization. Future studies can introduce some friction (e.g, longer forms) to increase participation cost. Furthermore, sending them earlier on a weekday would prevent bimodality, resulting in more homogeneous response windows.

We also realized the importance of having a good estimate of the data variance. Larger samples are required to study a diverse group. Additionally, focusing on a more culturally homogeneous target group would reduce the generalization of the results but increase the power of the tests.

Finally, due to the nature of the data distribution, other different approaches, like survival analysis (Parmet et al., 2014), could be used as they are designed for time-to-event data, handling right skewed distributions.

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Word Count: 2002 words.

Supplementary Information

A. Google Forms

Daily Preferences Mini Survey

Thank you for taking part in this short survey. We are Data Science and Machine Learning students conducting a small research project for one of our coursework assignments.

The aim of this study is to examine how everyday preferences and habits may vary across different cultures and countries.

The survey includes general questions about daily routines and preferences. The survey takes about one minute to complete, and your response will help us gather data for our study.

We truly appreciate your time and participation!

Question 1

Participant ID (Please Don't Change)

Question 2

How many glasses of water do you usually drink per day? {0-2,3-5,6-8,More than 8}

Question 3

How many hours of sleep do you get on an average night? {Less than 4 hours, 4-6 hours, 6-8 hours, More than 8 hours}

Question 4

Which do you prefer? {Tea, Coffee, Both, Neither}

Question 5

What is your favorite season? {Spring, Summer, Autumn, Winter}

Question 6

How many hours do you typically spend on your phone per day? {Less than 1 hour, 1-3 hours, 3-5 hours, More than 5 hours}

Question 7

Which type of music do you listen to most often? {Pop, Rock, Hip-Hop/Rap, Classical, Mixed/Everything}

Question 8

Which social media platform do you use the most? {Instagram, TikTok, Youtube, Snapchat, X, I don't use social media, Other}

Question 9

Which mode of transport do you use most frequently?{Walking, Public Transport, Car, Bicycle, Other}

Question 10

What king of breakfast do you usually prefer? {Eggs, Cereal/Muesli, Pastry, I don't usually eat breakfast, Other}

Question 11

What is your favorite type of movie? {Comedy, Action, Drama, Horror, Science Fiction, I don't watch movies}

B. Invitation Messages

- **Basic Message:** "Hi, I am currently collecting data for an MSc project and I am looking for participants to fill out a google forms. It takes less than 1 minute. Could you please fill it out at this link?"
- **Personalized Message:** "Hey [Name]. Quick favor—I'm working on my MSc project right now and I really need your help gathering some data. I want to understand how certain habits change depending on nationality and culture, and I really need a diverse range of people to make the results meaningful. The form is super short (literally less than a minute). Would you be able to help me out? Thanks!"